

1. Overall Model Performance

The model was evaluated on **60 images**, consisting of **30 AI-generated images** and **30 real images**.

Metric	Result
Images Tested	60
Correct Predictions	33
Incorrect Predictions	27
Accuracy	55.00%
Balanced Accuracy	55.00%
Precision (AI)	54.05%
Recall / Sensitivity (AI)	66.67%
Specificity (Real)	43.33%
F1-Score (AI)	59.70%

2. Confusion Matrix

The confusion matrix below shows how the model classified the 60 images:

Actual \ Predicted	AI	Real
Actual AI	20	10
Actual Real	17	13

This means:

- **True Positive (TP) = 20** → AI images correctly identified as AI.
- **False Negative (FN) = 10** → AI images incorrectly classified as real.
- **False Positive (FP) = 17** → Real images incorrectly classified as AI.
- **True Negative (TN) = 13** → Real images correctly identified as real.

3. AI Image Detection Performance

The model performed better at identifying **AI-generated images** than at identifying real images.

Recall / Sensitivity: 66.67%

The model correctly identified **20 out of 30 AI-generated images**.

However, it incorrectly classified **10 AI images as real**.

Precision: 54.05%

Out of all images predicted as AI, only **54.05% were actually AI-generated**.

This is because the model also incorrectly labelled **17 real images as AI**.

F1-Score: 59.70%

The F1-score combines precision and recall. A score of **59.70%** indicates moderate performance in detecting AI-generated images.

4. Real Image Detection Performance

Specificity: 43.33%

The model correctly identified only **13 out of 30 real images** as real.

However, **17 real images were incorrectly classified as AI-generated**.

This is one of the main findings of the evaluation: the model appears to have a tendency to predict **“AI-generated”**.

5. Classification Report

Class	Precision	Recall	F1-Score	Support
AI	54.05%	66.67%	59.70%	30
Real	56.52%	43.33%	49.06%	30
Accuracy			55.00%	60
Macro Average	55.29%	55.00%	54.38%	60
Weighted Average	55.29%	55.00%	54.38%	60

6. Overall Interpretation

The model achieved an overall **accuracy of 55.00%**, correctly classifying **33 out of 60 images**. Since the dataset was perfectly balanced, with 30 AI-generated and 30 real images, the **balanced accuracy was also 55.00%**.

The model demonstrated higher performance in detecting AI-generated images, achieving a **recall of 66.67%**, compared with a **specificity of 43.33%** for real images. This indicates that the model was more successful at identifying AI-generated images but frequently misclassified real images as AI-generated.

A total of **17 out of 30 real images** were incorrectly classified as AI, which suggests a bias towards predicting the AI class. Overall, the results indicate that the model achieved **moderate but limited performance** on the 60-image test set.

Short Research-Paper Summary

The model was evaluated on a balanced dataset of 60 images, comprising 30 AI-generated and 30 real images. It achieved an overall accuracy and balanced accuracy of 55.00%, correctly classifying 33 images. The model achieved a precision of 54.05%, recall of 66.67%, specificity of 43.33%, and F1-score of 59.70% for the AI class. The confusion matrix showed that 20 AI-generated images and 13 real images were correctly classified, while 17 real images were incorrectly predicted as AI. These results suggest that the model showed a tendency to classify images as AI-generated and was more effective at identifying AI images than real images.

